Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and
Communications Technology Department

Academic Year 2020 – 2021
Semester Spring
Exam Date 27/ 6 / 2021

Subject: (ELC 422) Digital Signal Processing
Examiners: Dr. Samir M Kamal & Dr. Sara M Hassan

QUESTIONS FOR THE FINAL WRITTEN EXAM

Spec.: Level 4 Elec. Eng. & Comm. Tech. Dep
Time: 2 hours

Number of Pages: One Number of Questions: 4 Attempt all Questions

Question 1 (20 points)

a- In sampling process, derive the sampling condition.
b- Prove that the DFT, X(K), of a sequence x(n) is periodic.
c- Check the periodicity of the following signals and for the periodic one, find its fundamental period:
(i) $x(n) = \cos(3n\pi)$ (ii) $x_a(t) = 3\cos(5t + \pi/6)$
d- For the following signal, x(n) = {1, 0, 1, 1}:
(i) Using the DFT method find the frequency components, X(k), and check your results by calculating x(n) using the IFFT algorithm.
(ii) Using your results, verify Parseval's theorem.
(iii) Plot the amplitude and the phase spectrums.

Question 2 (15 points)

a- What are the different steps of the digital filter design?
b- Compare between both filter types and list five advantages of the digital filter.
c- For the following two sequences, $x_1(n) = \{2, 2, 0, 2\}$ & $x_2(n) = \{1, 0, 1, 1\}$, using the direct methods, calculate:
(i) Their linear cross-correlation, then plot the obtained correlation function against the lag number, j.
(ii) The auto-correlation of the first sequence.
(iii) Their convolution.

Question 3 (20 points)

a- For LTIS define the following items: Linearity, Stability and ROC
b- Find, using the power series (recursive formula method) the first four values of the impulse response, h(n), of the system characterized by the following transfer function, H(z).

$$H(z) = \frac{(z+1)}{(z-0.5)(z-1)^2}$$

c- For the transfer function given in part (b), H(z):
(i) Plot its pole – zero diagram,
(ii) Find its region of convergence,
(iii) Discuss the stability of this system,
(iv) Find the difference equation of this system, y (n) and
(v) Draw its realization diagram using the block diagram method.

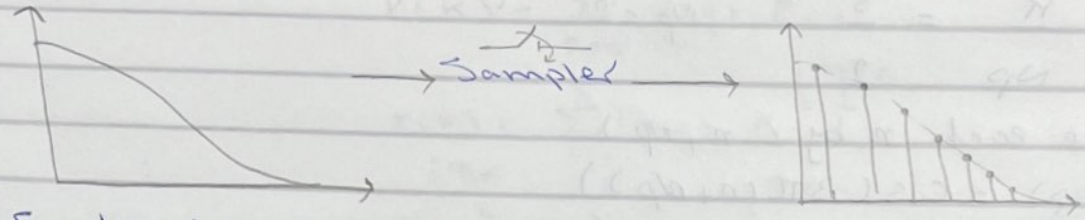
Question 4 (5 points)

Check (✓) for the right statements and (X) for the wrong statements and correct the mistakes.
a- In quantization process, when the quantization step (Δ) increases, the accuracy of the quantizer ( )
b- Nyquist frequency (F_N), is the minimum sampling frequency ( )
c- Fourier transform (FT) is calculated for a non-periodic continuous – time signals. ( )
d- Saving in the number of additions of N-point FFT is [N(N–2)–N $\log_2$ N] when compared with DFT operation ( )
e- To reconstruct the continuous time signal, x (t), from its discrete time sequence, X(n), the sampling rate, F_s, must be less than the frequency of x(t) signal, f_0. ( )

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Q1 :- a) Sampling process :- Converts Continuous time signal into discrete time signal.



$F \leq 1/2$ (cycle/sample)

$F = f/f_s$

$\frac{F}{f_s} \leq \frac{1}{2} \rightarrow F \leq \frac{f_s}{2} \rightarrow f_s \geq 2F$ 'Nyquist Condition'

Nyquist Rate :- minimum sampling rate to sample the signal without aliasing. $f_s = 2f_{max}$

b) $X(K) = \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi \frac{nK}{N}}$

$X(K) = X(K+N)$

$X(K+N) = \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi \frac{n(K+N)}{N}}$

$= \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi \frac{nK}{N}} \cdot e^{-j2\pi n}$ ($e^{-j2\pi n} = 1$)

$X(K+N) = \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi \frac{nK}{N}} = X(K)$

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c) i) $x(n) = \cos(3n\pi)$

$x(n) = A\cos(2\pi fn + \theta^\circ)$

$f = \frac{K}{Np} = \frac{3}{2}$, $Np = 2$

$2\pi f n = 3\pi K$

$2f = 3 \rightarrow f = \frac{3}{2}$

Replace each n by $(n + Np)$

$x(n+Np) = \cos(3\pi(n+Np))$

$= \cos(3\pi n + 6\pi)$

$x(n+Np) = \cos(3\pi n) = x(n)$

Signal is periodic every $Np = 2$ samples.

ii) $x_a(t) = 3\cos(5t + \pi/6)$

$x(t) = 3\cos(2\pi ft + \theta^\circ)$

$5t = 2\pi ft$

$5 = 2\pi f \rightarrow f = \frac{5}{2\pi}$

$Tp = \frac{1}{f} = \frac{2\pi}{5}$

Replace each t by $t + tp$

$x(t+tp) = 3\cos(5(t+tp) + \pi/6)$

$= 3\cos(5(t + \frac{2\pi}{5}) + \pi/6)$

$= 3\cos(5t + 2\pi + \pi/6)$

$x(t+tp) = 3\cos(5t + \pi/6) = x_a(t)$

Signal is periodic every $Tp = \frac{2\pi}{5}$ sec

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d) $x(n) = \{1, 0, 1, 1\}$

i) By using DFT :-

$$X(K) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi nK}{N}} \qquad N = 4$$

$$X(K) = \sum_{n=0}^{3} x(n)\, e^{-j\frac{\pi Kn}{2}}$$

$$X(K) = x(0) + x(1)\, e^{-j\frac{\pi K}{2}} + x(2)\, e^{-j\pi K} + x(3)\, e^{-j\frac{\pi 3K}{2}}$$

$$X(K) = 1 + e^{-j\pi K} + e^{-j\frac{\pi 3K}{2}}$$

At $(K) = 0 \rightarrow 3$

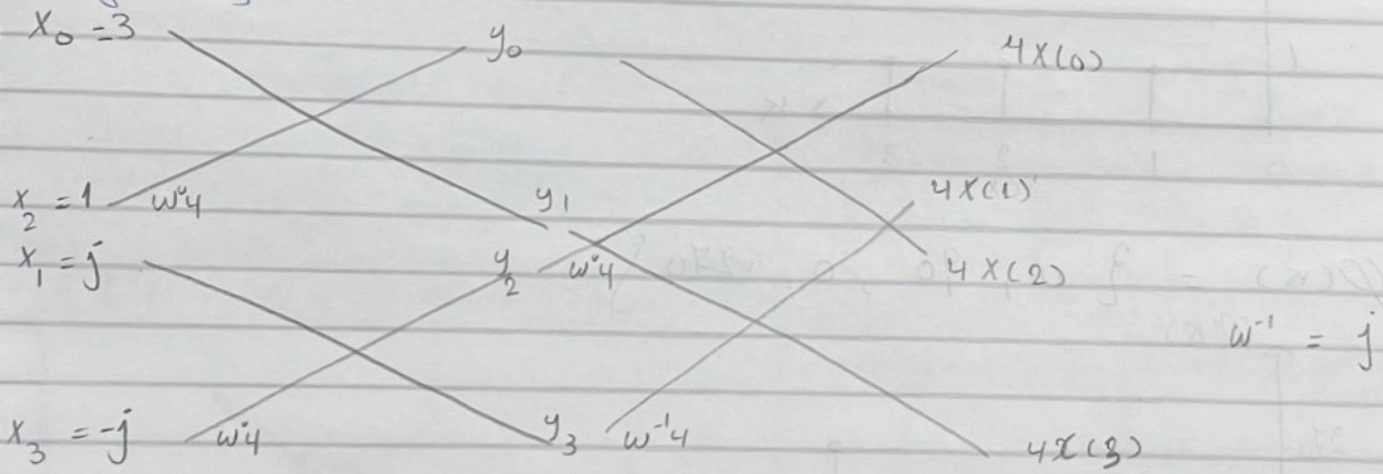
At $(K) = 1 \rightarrow +j$

At $(K) = 2 \rightarrow 1$

At $(K) = 3 \rightarrow -j$

$$X(K) = \{3, j, 1, -j\}$$

By using IFFT :-



$$Y_0 = X_0 + X_2 = 3 + 1 = 4$$
$$Y_1 = X_0 - X_2 = 3 - 1 = 2$$
$$Y_2 = X_1 + X_3 = j - j = 0$$
$$Y_3 = X_1 - X_3 = j + j = 2j$$

$$4x(0) = Y_0 + Y_2 = 4$$
$$4x(1) = Y_1 + jY_3 = 2 + j(2j) = 0$$
$$4x(2) = Y_0 - Y_2 = 4$$
$$4x(3) = Y_1 - jY_3 = 2 - j(2j) = 4$$

$$x(n) = \{1, 0, 1, 1\}$$

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ii) By parseval :-

$$X(K) = \{3, j, 1, -j\} \quad , \quad x(n) = \{1, 0, 1, 1\}$$

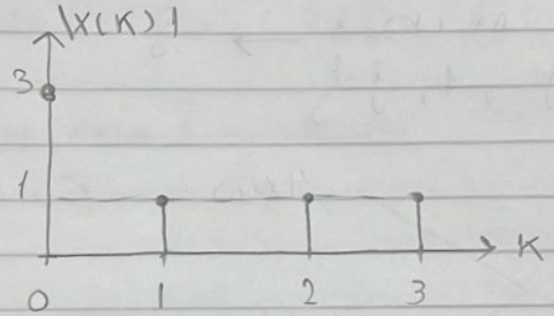
$$|X(K)| = \{3, 1, 1, 1\}$$

$$\sum_{n=0}^{N-1} x^2(n) = \frac{1}{4} \sum_{K=0}^{N-1} |X(K)|^2$$

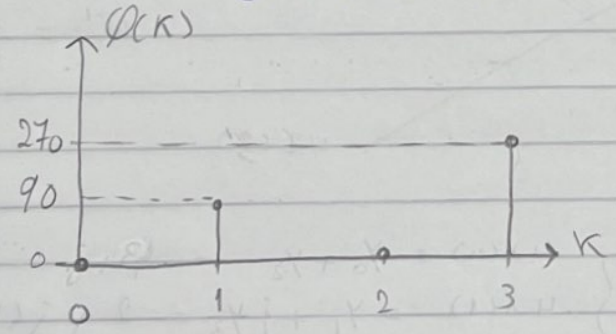
$$3 = 1/4\ (12) \rightarrow 3 = 3$$

parseval is verified.

iii)



$$\varphi(K) = \{0, 90, 0, +270\}$$



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Q2 :- a) 1- specification of filter's requirements.
2- calculation of filter's coefficients
3- Realization structure.
4- Analysis of effects of finite word length on filter performance.
5- implementation of filter in H/w &/or S/w + testing.

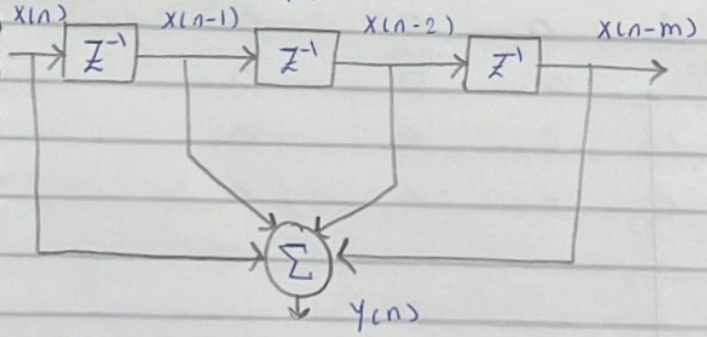
b) * Advantage of digital filters :-
1- has linear phase.
2- Doesn't vary with environmental change & thermal variations.
3- freq. response can be automatically adjusted if it is implemented by programmable processor.
4- filtered & unfiltered data can be saved for further use.
5- Can be used at very low freq & found in many biomedical apps.

* Comparison :-

FIR

- finite impulse response

$$H(z) = \sum_{k=0}^{N-1} h(k) z^{-k}$$



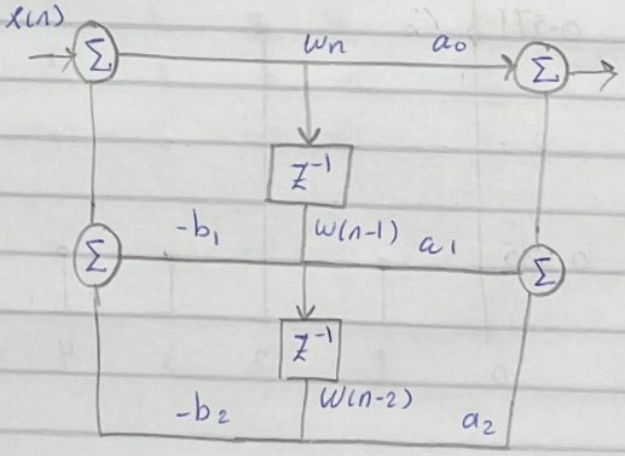
Diff eq:- $y(n) = h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) + \dots + h(m)x(n-m)$

how Delay & Accuracy
Always Stable 'No FB'
Realization

IIR

- infinite impulse response

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$



$w(n) = x(n) - b_1 w(n-1) - b_2 w(n-2)$

$y(n) = a_0 w(n) + a_1 w(n-1) + a_2 w(n-2)$

High Delay & Accuracy
May be stable may be not
Realization

← Lattice →

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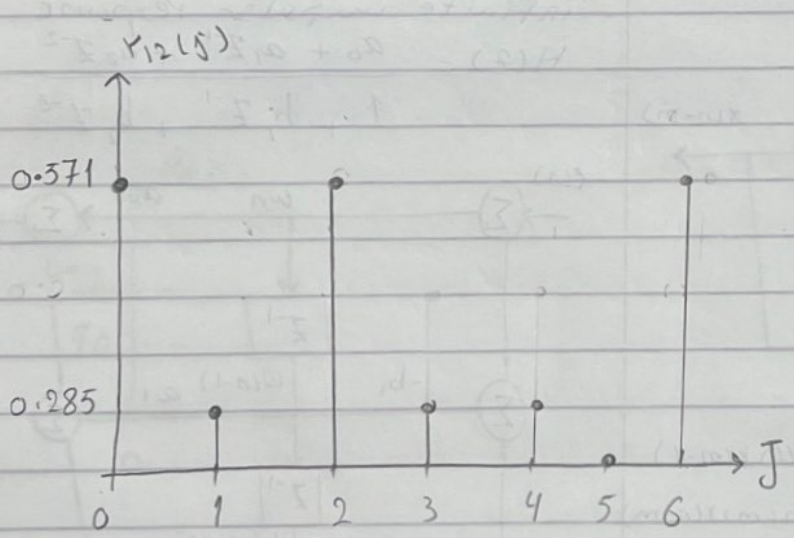
c) $x_1(n) = \{2, 2, 0, 2\}$ $\quad x_2(n) = \{1, 0, 1, 1\}$

i) By using Linear cross correlation

$N_1 = 4$, $N_2 = 4$, $N_{new} = 4 + 4 - 1 = 7$

									j	$r_{12}(j)$	$\rho_{12}(j)$
$x_1(n)$	2	2	0	2	0	0	0				
	1	0	1	1	0	0	0		0	0.571	0.666
	①	1	1	0	0	0	1		1	0.285	0.332
	1	1	0	0	0	1	0		2	0.571	0.666
$x_1(n+j)$	1	0	0	0	1	0	1		③	0.285	0.332
	0	0	0	①	0	1	1		4	0.285	0.332
	0	0	1	0	1	1	0		5	0	0
	0	1	0	1	1	0	0		6	0.571	0.666

$$\rho_{12} = \frac{r_{12}(j)}{\frac{1}{N}\sqrt{\sum_{0}^{N_{new}-1} x_1^2(n) \sum_{0}^{N_{new}-1} x_2^2(n)}} = \frac{r_{12}(j)}{0.857}$$



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ii) $x_1(n) = \{2, 2, 0, 2\}$ $N_{new} = 4$

$x_1(n)$					$r_{11}(j)$	$\rho_{11}(j)$
2	2	0	2	0	3	1
2	2	0	2	0		
2	0	2	0	1	1	0.333
0	2	0	0	2	1	0.333
2	0	0	0	3	1	0.333

$$\rho_{11}(j) = \frac{r_{11}(j)}{\frac{1}{4}\frac{1}{N}\sum_{0}^{N_{new}-1} x_1^2(n)} = \frac{r_{11}(j)}{3}$$

iii) Convolution $x_1(n) = \{2, 2, 0, 2\} \rightarrow \{0, 0, 0, 2, 2, 0, 2\}$

$x_2(n) = \{1, 0, 1, 1\} \rightarrow \{1, 1, 0, 1, 0, 0, 0\}$

$x(n)$								$y(n)$
0	0	0	2	2	0	2	n	
1	1	0	1	0	0	0	0	2
0	1	1	0	1	0	0	1	2
0	0	1	1	0	1	0	2	2
0	0	0	1	1	0	1	3	6
0	0	0	0	1	1	0	4	2
0	0	0	0	0	1	1	5	2
0	0	0	0	0	0	1	6	2

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Q3 :- a) Linearity :- A discrete time system is said to be linear if it obeys the principle of superposition

$$a_1 x_1(n) + a_2 x_2(n) \longrightarrow a_1 y_1(n) + a_2 y_2(n)$$

Stability : LTIs is stable if its impulse response satisfy the following condition. $y(n) = \sum_{k=-\infty}^{\infty} h(k) < \infty$

ROC :- Region of Convergence, all the values that make H(z) finite [Roc<∞

b) $$H(z) = \frac{z+1}{z^3 - 2z^2 + z - 0.5z^2 + z - 0.5} = \frac{z+1}{z^3 - 2.5z^2 + 2z - 0.5} * \frac{z^{-3}}{z^{-3}}$$

$$H(z) = \frac{z^{-2} + z^{-3}}{1 - 2.5z^{-1} + 2z^{-2} - 0.5z^{-3}} = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3}}{b_0 + b_1 z^{-1} + b_2 z^{-2} + b_3 z^{-3}}$$

$a_0 = 0$, $a_1 = 0$, $a_2 = 1$, $a_3 = 1$, $a_4 \to a_\infty = 0$

$b_0 = 1$, $b_1 = -2.5$, $b_2 = 2$, $b_3 = -0.5$, $b_4 \to b_\infty = 0$

$$h(n) = \left[a_n - \sum_{i=1}^{n} h(n-i) b_i\right] / b_0$$

$h(0) = a_0/b_0 = 0$

$h(1) = [a_1 - h(0)b_1]/b_0 = 0 - 0/1 = 0$

$h(2) = [a_2 - h(1)b_1 - h(0)b_2]/b_0 = [1 - 0 - 0]/1 = 1$

$h(3) = [a_3 - h(2)b_1 - h(1)b_2 - h(0)b_3]/b_0 = [1 + 2.5 - 0.0]/1 = 3.5$

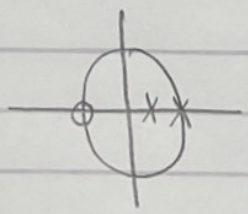
i) poles → $z = \{0.5, 1, 1\}$, Zeros → $z = \{1\}$

ii) Roc is the entire z-plane except $z = \{0.5, 1\}$

o → Zeros

x → poles

iii) System is marginally stable as there's one pole on the unit circuit & no poles outside.



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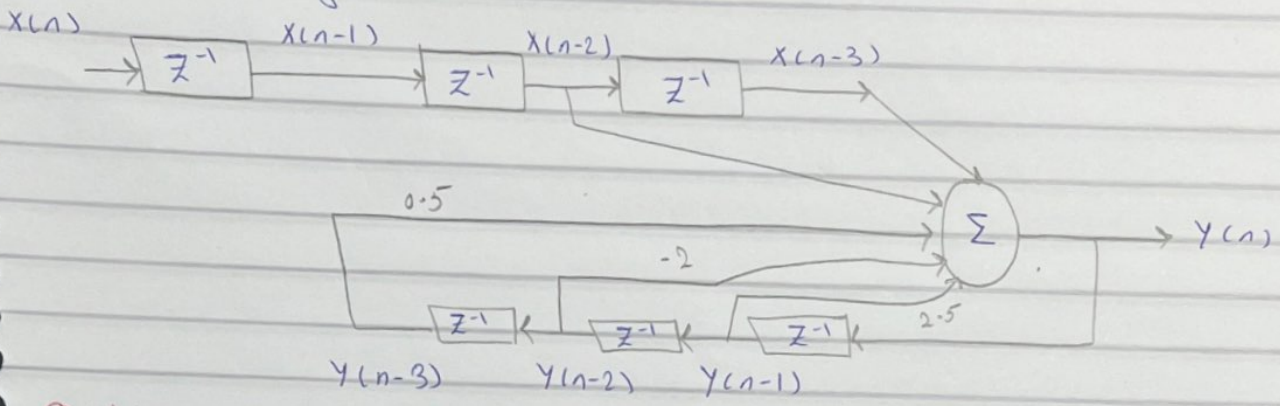
iv) $H(z) = \dfrac{z^{-2} + z^{-3}}{1 - 2.5z^{-1} + 2z^{-2} - 0.5z^{-3}} = \dfrac{Y(z)}{X(z)}$

$Y(z) - 2.5Y(z)z^{-1} + 2Y(z)z^{-2} - 0.5Y(z)z^{-3} = X(z)z^{-2} + X(z)z^{-3}$

$Y(z) = X(z)z^{-2} + X(z)z^{-3} + 2.5Y(z)z^{-1} - 2Y(z)z^{-2} + 0.5Y(z)z^{-3}$

$y(n) = x(n-2) + x(n-3) + 2.5y(n-1) - 2y(n-2) + 0.5y(n-3)$

v) Block Diagram



Q4 :-

a) X → Decrease

b) ✓

c) ✓

d) X → $N\log_2 N - N(N-2)$

e) X → morethan

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